

PSLE MATHEMATICS PRACTICE PAPER 8

Coordinates & Graphs

Duration: 1 hour

Total Marks: 50

Name: _____

Date: _____

SECTION A: Short Answer Questions (20 marks)

1. Plot point A(3, 4) on a coordinate grid. (2 marks)

2. What are the coordinates of the point marked B if it is 5 units right and 3 units up from the origin? (2 marks)

3. Find the distance between points A(0, 2) and P(3, 4) on a grid. (2 marks)

4. Write the equation of a line passing through (0, 0) and (1, 2). (2 marks)

5. If $y = 2x$, find y when $x = 5$. (2 marks)

6. What is the y-intercept of the line $y = 2x + 3$? (2 marks)

7. On a graph, $y = 3$. What type of line is this? (2 marks)

8. Find the midpoint between (2, 4) and (8, 10). (3 marks)

SECTION B: Long Answer Questions (30 marks)

9. Plot the line $y = x + 2$ on a grid from $x = -2$ to $x = 4$. (4 marks)

10. The vertices of a square are $A(0,0)$, $B(4,0)$, $C(4,4)$, $D(0,4)$. Find its perimeter and area. (3 marks)

11. A line passes through points $(1, 3)$ and $(3, 7)$. Find the equation of the line. (4 marks)

12. Plot $y = 2x - 1$ and $y = x + 1$ on the same grid. Find their intersection point. (4 marks)

13. A triangle has vertices $A(0, 0)$, $B(6, 0)$, $C(3, 6)$. Find its area. (3 marks)

14. Reflect point $P(3, 2)$ across the y-axis. Write the coordinates of P' . (2 marks)

15. A line has gradient 2 and passes through $(0, 5)$. Write its equation and find where it crosses the x-axis. (4 marks)

SOLUTIONS

SECTION A SOLUTIONS

Q1: Plot A(3,4)

Mark point 3 units right, 4 units up

Q2: Point B coordinates

B(5, 3)

Q3: Distance A to B

Distance = $\sqrt{3^2 + 4^2} = \sqrt{25} = 5$

Q4: Line equation through origin and (1,2)

$y = 2x$

Q5: $y = 2x$ when $x = 5$

$y = 2(5) = 10$

Q6: y-intercept of $y = 2x + 3$

y-intercept = 3 (when $x = 0$)

Q7: $y = 3$

Horizontal line parallel to x-axis

Q8: Midpoint (2,4) to (8,10)

Midpoint = $((2+8)/2, (4+10)/2) = (5, 7)$

SECTION B SOLUTIONS

Q9: Line $y = x + 2$

Plot line passing through (-2, 0), (-1, 1), (0, 2), (1, 3), (2, 4), (3, 5), (4, 6)

Q10: Square perimeter and area

Perimeter = $4 \times 4 = 16$, Area = $4 \times 4 = 16$

Q11: Line through (1,3) and (3,7)

Gradient = $(7-3)/(3-1) = 2$, $y = 2x + 1$

Q12: Intersection of $y = 2x-1$ and $y = x+1$

$2x - 1 = x + 1$, $x = 2$, $y = 3$. Intersection: (2, 3)

Q13: Triangle area

$$\text{Area} = \frac{1}{2} \times \text{base} \times \text{height} = \frac{1}{2} \times 6 \times 6 = 18$$

Q14: Reflect P(3,2) across y-axis

$$P'(-3, 2)$$

Q15: Line equation

$$y = 2x + 5. \text{ x-intercept: } 0 = 2x + 5, x = -2.5$$

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